CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS :VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                              struct nighttime {
      {                                            int  x, y,z;
      int hr,min,sec;                                    }
      }
            struct {
         int hr,min,sec:
               } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

5

b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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* learning curve

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CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

Answer any 5 PART – A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART – B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
   c) Explain the approaches used in defining the semantics? **(3)**
8. a) What is the support for streams for sequential I/O in java. **(5)**
   b) Define an exception. Give five examples for excempions at different levels of abstraction **(3+5)**
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area. **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

Part-C

14/14

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B**. (2+4)

2 4

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: **(4+4)**

14 8

```
// compute the factorial of n
void main ( ) {
   int n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS :VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime
    {
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struct nighttime {
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PART – B (2 X 13 = 26 Marks) Answer any 2

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}
function1();
function2();
function3();
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PART – C (14 marks)

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

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Answer any 5 PART - A (5 X 2 = 10) 2023103062

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2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
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5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
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9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
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```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

Part-C

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.**
(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
 }
}
```

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ANNA UNIVERSITY, CHENNAI - 600025
BE-CSE VI SEM - N, P & Q Batches

Date: 02-02-2026 **Max Marks: 50**
Time:1.30 mins

PART A (10*2=20 Marks)

1. What are the components of a compiler front end?
2. Compare compiler vs interpreter with respect to execution.
3. Consider the context-free grammar
   S ->S S +| S S *| a
   Show how the string **aa+a*** can be generated by this grammar.
4. Write a Regular Expression for a valid email ID (basic format).
5. What language is generated by the following grammar?
   S -> a| S + S| S S| S *| (S)
6. What is panic mode error recovery in lexical analysis?
7. Remove left recursion in the following grammar
   S → A B
   A → Aa | ε
   B → Bb | c
8. Define Lexeme and Pattern.
9. Differentiate Buffer pairs and Sentinels.
10. Consider the context-free grammar
    E → E + E | E * E | id
    Perform handle pruning for the string **id + id + id**.

PART B (30 Marks)

ANSWER ALL QUESTIONS

11. a. Is grammar LL(1)? (10)
    S→A a | b
    A→A c | S d | ε
    (OR)
    b. Is grammar LL(1)?
    S→ i E tS | i E t S e S | a
    E→ b
12. Show that the grammar is CLR(1) but not SLR: (15)
    S→ L = R | R
    L→*R | id
    R→L
13. For the above grammar (Q.No. 12) construct LALR parsing table. (5)

*******ALL THE BEST*******

S → A a | b
A → S d A' | A'
A' → c A' | ε

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ANNA UNIVERSITY, CHENNAI - 600025

BE-CSE VI SEM – N & P Batches

Assessment II : CS23602-COMPILER DESIGN

Date: 26-03-2026 **Max Marks: 50**

Time:1.30 mins

PART A (10*2=20 Marks)

PART B (30 Marks)

ANSWER ALL QUESTIONS

1. a. Attributes computed using only child nodes are called ________ attributes.
   b. A parse tree where each node has computed attribute values is called _____
2. Write short circuit code for if (a==b || c==d || e==f) x = 1
3. Write semantic action for type conversions into expression evaluation.
4. Write the semantic rules for constructing syntax tree for simple expressions.
5. Write the type expressions for array, function, pointer and record.
6. Write SDT for repeat-until (do-while) loop structure
7. Translate the expression (a*-b)+e^2 to 3 address code in Triples
8. Compute liveness and next-use information for the following basic block.

a=b+c
d=a+b

9. Eliminate the local common sub expression in the following basic block using DAG

1) t1 = a + b
2) t2 = c - d
3) t3 = a + b
4) t4 = t3 * e
5) t5 = c - d
6) t6 = t5 + t4

10. What is the significance of next-use information in register allocation?

PART B (3*10=30 Marks)

Answer any 3 Questions

11. Write the SDT rules and draw the annotated parse tree with relevant SDT scheme for the following statement.

A = B[i, j] + D[k]

Integer values held at B[i,j]=45, D[i]=78, D[k]=24, Sizes of the arrays are: B =40x40, D =50, i=j=k=25, assume base addresses.

12. Store the ICG form of the high level code in quadruple where set is a function with arguments and p is a single dimension array

S = set (a, 9)

A = S + p[i]

13. Using Backpatching of Boolean Expression and control statements translation technique convert the following if condition to 3-address code. Assume start address at 100. Draw the parse tree.

```
a = - c * d
if (((a > b) && ! (c = = d)) || (g))
        a = f * a
else
    a = b++
```

14. Construct the flow graph for the following three address code, identify the loops in the flow graph and identify local common sub expressions (if any) for each loop.

```
1) i = 1
2) j = 1
3) t1 = 10 * i
4) t2 = t1 + j
5) t3 = 8 * t2
6) t4 = t3 - 88
7) c[t4] = 0.0
8) k = 1
9) t5 = 10 * i
10) t6 = t5 + k
11) t7 = 8 * t6
12) t8 = t7 - 88
13) t9 = a[t8]
14) t10 = 10 * k
15) t11 = t10 + j
16) t12 = 8 * t11
17) t13 = t12 - 88
18) t14 = b[t13]
19) t15 = t9 * t14
20) t16 = 10 * i
21) t17 = t16 + j
22) t18 = 8 * t17
23) t19 = t18 - 88
24) t20 = c[t19]
25) t21 = t20 + t15
26) c[t19] = t21
27) k = k + 1
28) if k <= 10 goto (9)
29) j = j + 1
30) if j <= 10 goto (3)
31) i = i + 1
32) if i <= 10 goto (2)
```